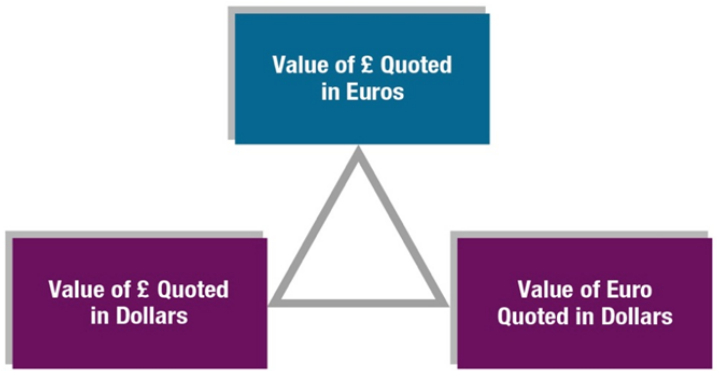
Exhibit 7.8 Comparing Arbitrage Strategies

Locational Arbitrage: Capitalizes on discrepancies in exchange rates across locations.



Triangular Arbitrage: Capitalizes on discrepancies in cross exchange rates.



Covered Interest Arbitrage: Capitalizes on discrepancies between the forward rate and the interest rate differential.



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Interest Rate Parity (3 of 6)

- **Determining the Forward Premium**
- **Effect of the interest rate differential:** The relationship between the forward premium (or discount) and the interest rate differential according to IRP is simplified in an approximated form:

- where
- p = forward premium (or discount)
- F = forward rate in dollars
- S = spot rate in dollars
- i_h = home interest rate
- i_f = foreign interest rate

$$F = S \cdot (1 + p) \Rightarrow p = \frac{F}{S} - 1$$

$$p = i_h - i_f$$

- **Implications:** If the forward premium is equal to the interest rate differential as just described, then covered interest arbitrage will not be feasible.

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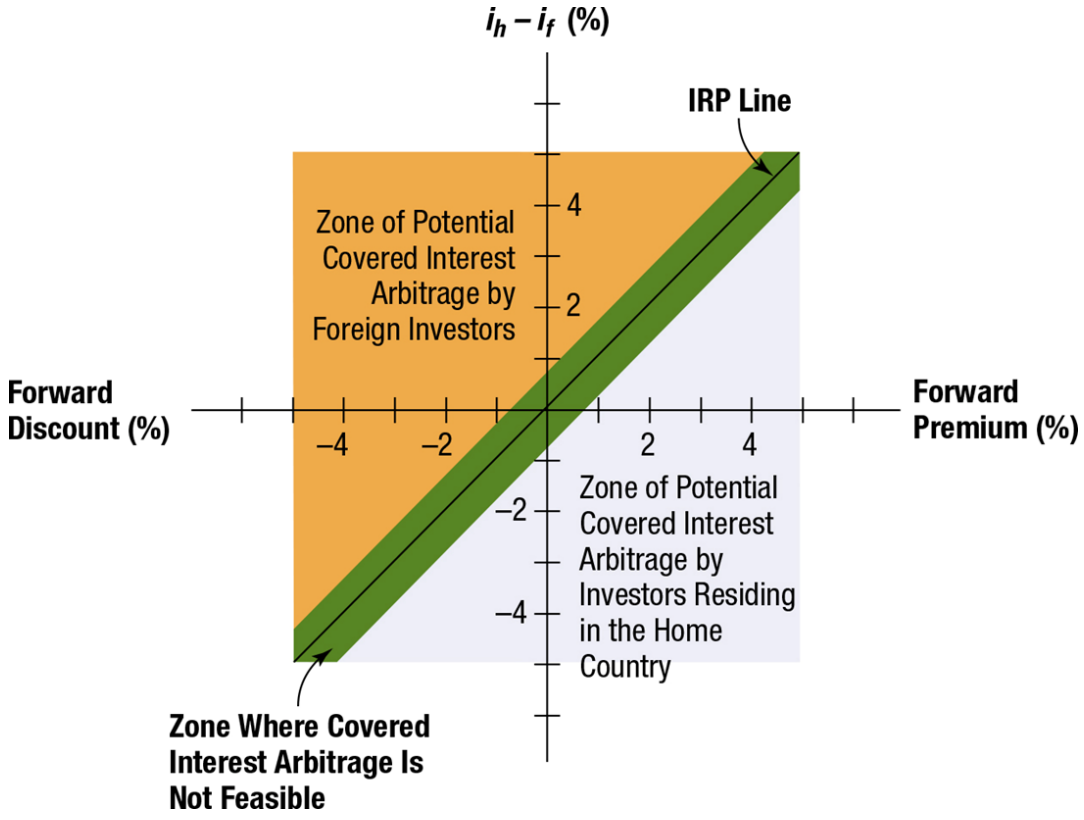
Covered Interest Rate Arbitrage

takes advantage of misalignment between interest rates and forward and spot rates while insulating against exchange rate movements with a futures/forward contract.

Arbitrage opportunity exists $P \neq i_h - i_f$

$P > i_h - i_f \rightarrow$ Arbitrage opportunity at home

Exhibit 7.10 Potential for Covered Interest Arbitrage When Considering Transaction Costs



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Interest Rate Parity (6 of 6)

Considerations When Assessing Interest Rate Parity

- **Transaction costs**
  - The actual point reflecting the interest rate differential and forward rate premium must be farther from the IRP line to make covered interest arbitrage worthwhile. (See Exhibit 7.10)
- **Political risk**
  - A crisis in the foreign country could cause its government to restrict any exchange of the local currency for other currencies.
- **Differential tax laws**
  - Covered interest arbitrage might be feasible when considering before-tax returns but not necessarily when considering after-tax returns.

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$F = i_h - i_f$ → Interest Rate Parity Holds

$F > i_h - i_f$ → arbitrage opportunity

→

t=0 Borrow funds at home

Convert at spot rate S

Deposit in foreign account

t=0 Sell futures contract/ enter into futures contract to sell foreign currency

t=1 -> Take out funds deposited abroad

-> pay back loan locally and close out futures contract

$P < i_h - i_f$ — arbitrage opportunity in a foreign

t=0 Borrow funds abroad, sell FX/buy USD deposit in USA in spot market

t=0. Open a futures contract to BUY EX in futures market

t=1 Take out your USD deposit, close out futures position, buy back EX payback the loan.

90 day futures and interest rates are as follows:

$\$800{,}000$

UK_{90} 4%

US_{90} 2%

$F_{90} = \$1.60/£$

$S = \$1.60/£$

$$\frac{F}{S} - 1 = p = i_h - i_f$$

$$\frac{\$1.60}{\$1.60} - 1 = 1 - 1 = 0$$

$$2\% - 4\% = -2$$

$p > -2\% \Rightarrow$ the arbitrage opportunity exists

t: 0

Borrow $800,000 at $i_h = 2\%$

Convert $800,000 to GBP: at Spot buy GBP / sell USD

enter into futures contract to SELL GBP for the amount you will take out

Deposit in British bank at $i_f = 4\%$

$800,000 / S = $1.60/£

£500,000

$t=1$ Withdraw

£509000 · (1+4%)

= £529000

Sell GBP/Buy USD

-> deliver the futures contract => sell at £=$1.60/£

£529000 · $1.60/£

= $832,000

(You can also convert GBP in spot market and offset futures contract at current forward rate)

Pay Back the loan

$800,000 · (1+2%)

= $816,000

Riskless π = $832,000

− $816,000

$16,000